EXPERIMENT NO. 1

Title

Probability Distribution of Categorical Variables Using Python

Aim

To calculate the probability distribution of categorical variables such as **Gender, Batting Style, Bowling Style, Player Position, and Country** from the given dataset using Python and the Pandas library.

Problem Statement

From the given dataset **players_info.csv**, perform the following tasks:

a) Calculate the probability distribution of genders among the players.

b) Calculate the probability distribution of batting styles.

c) Calculate the probability distribution of bowling styles.

d) Calculate the probability distribution of player positions.

e) Calculate the probability distribution of countries among the players.

Learning Objectives

After completing this experiment, the student will be able to:

- Understand the concept of probability distribution.
- Read a CSV dataset using the Pandas library.
- Calculate the frequency of categorical variables.
- Compute probability values using mathematical formulas.
- Display the results in tabular format using Python.
- Verify that the sum of all probabilities equals 1.

Theory

Data Analytics involves collecting, organizing, analysing, and interpreting data to obtain useful information. One of the fundamental statistical concepts used in data analysis is the **probability distribution**.

A probability distribution shows how the total observations in a dataset are distributed among different categories. It indicates the likelihood of occurrence of each category.

For categorical data, the probability of a category is calculated using the following formula:

$$P(E) = \frac{\text{Frequency of the Category}}{\text{Total Number of Records}}$$

Where:

- **P(E)** represents the probability of an event.
- **Frequency** is the number of times a category appears in the dataset.
- **Total Number of Records** is the total number of observations present in the dataset.

The probability values always satisfy the following condition:

$$0 \leq P(E) \leq 1$$

The sum of the probabilities of all categories is always equal to:

$$\sum P(E) = 1$$

In Python, the **Pandas** library provides the `value_counts()` function to calculate the frequency of each category, making the computation of probability distributions simple and efficient.

Software Requirements

- Operating System: Windows 10/11 or Linux
- Python 3.x
- Anaconda Distribution
- Jupyter Notebook
- Pandas Library
- CSV Dataset: **players_info.csv**

Formula

The probability of each category is calculated using:

$$Probability = \frac{Frequency}{Total\ Number\ of\ Records}$$

Percentage is calculated using:

$$Percentage = Probability \times 100$$

Algorithm

Step 1: Start Jupyter Notebook using Anaconda Navigator.

Step 2: Create a new Python Notebook.

Step 3: Import the Pandas library.

Step 4: Read the dataset `players_info.csv` using the `read_csv()` function.

Step 5: Display the first five records of the dataset.

Step 6: Display the column names of the dataset.

Step 7: Find the total number of records in the dataset.

Step 8: Calculate the frequency of each category using the `value_counts()` function.

Step 9: Calculate the probability by dividing the frequency by the total number of records.

Step 10: Convert the probability into percentage.

Step 11: Display the results in tabular format.

Step 12: Verify that the sum of all probabilities is equal to 1.

Procedure

Step 1

Open **Anaconda Navigator**.

Step 2

Launch **Jupyter Notebook**.

Step 3

Create a new **Python 3 Notebook**.

Step 4

Save the notebook with the name:

Experiment_01_Probability_Distribution.ipynb

Step 5

Copy the dataset **players_info.csv** into the same folder where the notebook is saved.

Step 6

Import the Pandas library.

import pandas as pd

Step 7

Read the CSV dataset.

df = pd.read_csv("players_info.csv")

Step 8

Display the first five records.

print(df.head())

This step verifies that the dataset has been loaded successfully.

Step 9

Display the column names.

print(df.columns)

This helps identify the available columns in the dataset.

Step 10

Find the total number of records.

total_players = len(df)

This value is used as the denominator while calculating probabilities.

Step 11

Calculate the frequency of each category using the `value_counts()` function.

Example:

gender_frequency = df["gender"].value_counts()

The `value_counts()` function counts the number of occurrences of each category.

Step 12

Calculate the probability of each category.

Example:

gender_probability = gender_frequency / total_players

The probability is calculated by dividing the frequency by the total number of records.

Step 13

Convert the probability into percentage.

percentage = gender_probability * 100

Step 14

Display the frequency, probability, and percentage in a tabular format.

Step 15

Repeat Steps 11–14 for the remaining categorical variables:

- Batting Style
- Bowling Style
- Player Position
- Country

Step 16

Verify the probability distribution.

print(gender_probability.sum())

The output should be: 1.0

Similarly, verify the probability distribution for all the remaining categories.

Python Program

```
# Experiment 1
# Program to calculate the probability distribution
# of Gender, Batting Style, Bowling Style,
# Position and Country.

# Import the Pandas library
import pandas as pd

# Read the dataset
df = pd.read_csv("players_info.csv")
```

```
# Display the first five records
print("First Five Records")
print(df.head())

# Display the column names
print("\nColumn Names")
print(df.columns)

# Find the total number of players
total_players = len(df)

print("\nTotal Number of Players =", total_players)
# ---------------------------------------
# a) Probability Distribution of Gender
# ---------------------------------------
print("\na) Probability Distribution of Gender")
gender_frequency = df["gender"].value_counts()
gender_probability = gender_frequency / total_players
gender_result = pd.DataFrame({
    "Frequency": gender_frequency,
    "Probability": gender_probability,
    "Percentage": gender_probability * 100
})

print(gender_result)
print("\nTotal Probability =", gender_probability.sum())
```

```
# ---------------------------------------
# b) Probability Distribution of Batting Style
# ---------------------------------------
print("\nb) Probability Distribution of Batting Style")
batting_frequency = df["batting_style"].value_counts()
batting_probability = batting_frequency / total_players

batting_result = pd.DataFrame({
    "Frequency": batting_frequency,
    "Probability": batting_probability,
    "Percentage": batting_probability * 100
})
print(batting_result)
print("\nTotal Probability =", batting_probability.sum())
# ---------------------------------------
# c) Probability Distribution of Bowling Style
# ---------------------------------------
print("\nc) Probability Distribution of Bowling Style")
bowling_frequency = df["bowling_style"].value_counts()
bowling_probability = bowling_frequency / total_players

bowling_result = pd.DataFrame({
    "Frequency": bowling_frequency,
    "Probability": bowling_probability,
    "Percentage": bowling_probability * 100
})
```

```
print(bowling_result)
print("\nTotal Probability =", bowling_probability.sum())
# ---------------------------------------
# d) Probability Distribution of Position
# ---------------------------------------
print("\nd) Probability Distribution of Player Position")
position_frequency = df["position"].value_counts()
position_probability = position_frequency / total_players

position_result = pd.DataFrame({
    "Frequency": position_frequency,
    "Probability": position_probability,
    "Percentage": position_probability * 100
})
print(position_result)
print("\nTotal Probability =", position_probability.sum())
# ---------------------------------------
# e) Probability Distribution of Country
# ---------------------------------------
print("\ne) Probability Distribution of Country")
country_frequency = df["country"].value_counts()
country_probability = country_frequency / total_players
country_result = pd.DataFrame({
    "Frequency": country_frequency,
    "Probability": country_probability,
    "Percentage": country_probability * 100
})
```

```
print(country_result)

print("\nTotal Probability =",

    country_probability.sum())
```

Expected Output

The program should display:

- First five records of the dataset.
- Column names.
- Total number of records.
- Frequency table for each categorical variable.
- Probability table for each categorical variable.
- Percentage distribution.
- Sum of probabilities equal to 1.0.

Result

The probability distributions of Gender, Batting Style, Bowling Style, Player Position, and Country were successfully calculated from the `players_info.csv` dataset using Python and the Pandas library. The calculated probabilities were verified, and the sum of probabilities for each categorical variable was found to be equal to 1, confirming the correctness of the probability distributions.

```
First Five Records
  player_id   player_name gender batting_style   bowling_style      position  \
0      P001  Aarav Sharma   Male        Switch  Right-arm pace        Batter
1      P002   Rohan Singh   Male    Right-hand  Right-arm pace        Bowler
2      P003  Grace Sharma   Male     Left-hand  Right-arm spin        Batter
3      P004  Arjun Sharma   Male     Left-hand  Right-arm spin  Wicketkeeper
4      P005   Kabir Brown   Male    Right-hand  Right-arm pace        Batter

     country
0   Pakistan
1      India
2  Australia
3  Australia
4  Australia

Column Names
Index(['player_id', 'player_name', 'gender', 'batting_style', 'bowling_style',
       'position', 'country'],
      dtype='object')

Total Number of Players = 80

a) Probability Distribution of Gender
        Frequency  Probability  Percentage
gender
Male           56          0.7        70.0
Female         24          0.3        30.0

Total Probability = 1.0


b) Probability Distribution of Batting Style
               Frequency  Probability  Percentage
batting_style
Right-hand            48         0.60        60.0
Left-hand             28         0.35        35.0
Switch                 4         0.05         5.0

Total Probability = 1.0

c) Probability Distribution of Bowling Style
                   Frequency  Probability  Percentage
bowling_style
Right-arm pace            24        0.300        30.0
Right-arm spin            20        0.250        25.0
Left-arm pace             14        0.175        17.5
Left-arm spin             12        0.150        15.0
Wicketkeeper/None         10        0.125        12.5

Total Probability = 1.0

d) Probability Distribution of Player Position
              Frequency  Probability  Percentage
position
Batter               24        0.300        30.0
Bowler               22        0.275        27.5
All-rounder          20        0.250        25.0
Wicketkeeper         14        0.175        17.5

Total Probability = 1.0
```

e) Probability Distribution of Country

country	Frequency	Probability	Percentage
India	20	0.2500	25.00
Australia	12	0.1500	15.00
England	12	0.1500	15.00
South Africa	10	0.1250	12.50
New Zealand	8	0.1000	10.00
West Indies	7	0.0875	8.75
Pakistan	6	0.0750	7.50
Sri Lanka	5	0.0625	6.25

Total Probability = 1.0